Sourcing and loading data Activity Book

Data Modeling for Qlik Sense

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Activity 1: Create the ABC database connection

Summary:

- Create a data connection to an Azure SQL database.
- Test the connection.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, create a new connection with the following properties:

- Connection: **Azure SQL Database**
- Server: **qlik-es.database.windows.net**
- Port: **1433**
- Database: **ABC_2025**
- User name: **QlikEducationStudent**
- Password: **AzureTraining1**
- Name: **ABC**

Hints

- New connections can be created from the **Data load editor**.
- The property values need to be added manually.
- Testing the connection is always a wise choice.

Step-by-step guide

1. In the **Data load editor**, create a new data connection.
2. Select **Azure SQL Database** from the provider list.
3. Enter the **Server** value of: *qlik-es.database.windows.net*.
4. Ensure the **Port** is set to: *1433*.
5. Enter the **Database** name of: *ABC_2025*.
6. For the **User name**, use: *QlikEducationStudent*.
7. Provide a **Password** of: *AzureTraining1*.
8. Edit the default-generated **Name** (bottom most field in the pop-up window) to: *ABC*.
9. Test the connection and make sure it is successful.
10. **Create** the connection.
11. After you create the connection, confirm it appears in the **Data connections** panel.

> **Qlik Sense on Windows:** The connection name also contains the user that created the connection, **ABC (qse_student)**.

Solution

Activity 2: Load Orders data

Summary:

- Use the ABC data connection to work with the select data control.
- Generate the load/select script to extract all fields from the Orders table.
- Execute the script to bring data into the app

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, load a new table to start creating your data model. To achieve this, do the following:

- Navigate to the **Orders** section in the **Data load editor**.
- Use a data connection to select your data.
- Load all the fields from the *orders* table and include a preceding **LOAD** in your script.
- Name the table *Orders* and add a description.

Hints

- The **Select data** dialog in the data connection can be used to load the table.
- When a database is used as a data source, the "Include LOAD statement" check box needs to be selected to also add the Qlik Sense script.
- The name of the table always ends with a colon (:) and the comment syntax begins with two forward slashes (//) .

Step-by-step guide

1. In the **Data load editor**, navigate to the **Orders** section.
2. Use the correct data connection and click the **Select data** icon to display the dialog window.
3. Use the **Owner** drop-down to select *dbo* as the owner.
4. Select the *orders* table from the **ABC** database and make sure all the fields are selected before inserting the script.
5. Ensure that the "Include LOAD statement" box is checked.

6. **Insert script** and make sure it resembles the following:

```
LIB CONNECT TO 'ABC';

LOAD OrderID,
    CustomerID,
    EmployeeID,
    ShipperID,
    FreightWeight,
    OrderDate;

[orders]:
SELECT OrderID,
    CustomerID,
    EmployeeID,
    ShipperID,
    FreightWeight,
    OrderDate
FROM "ABC_2025".dbo.orders;
```

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Qlik Sense on Windows: The connection name also contains the user that created the connection, **ABC (qse_student)**.

7. Relocate the table label *[orders]:* from above the **SELECT** statement, to above the preceding **LOAD** statement and capitalize the "O".
8. Directly above the table label, add a comment to describe the table and make sure the script looks as follows:

```
LIB CONNECT TO 'ABC';

// **** Load Orders table ****
[Orders]:
LOAD OrderID,
    CustomerID,
    EmployeeID,
    ShipperID,
    FreightWeight,
    OrderDate;
SELECT OrderID,
    CustomerID,
    EmployeeID,
    ShipperID,
    FreightWeight,
    OrderDate
FROM "ABC_2025".dbo.orders;
```

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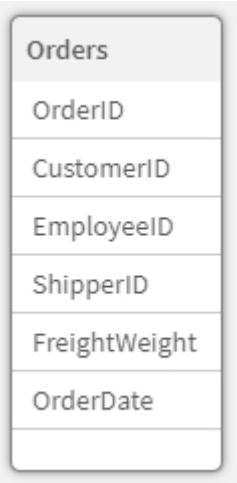
9. Click the **Load data** button and review the new table in the **Data model viewer**.

Solution



Activity 3 - Challenge: Load OrderDetails and Customers data

Summary:

- Use the existing data connection to select and load data from the orderdetails and customers tables.
- Load the orders table data and verify it is associated with orderdetails and customers tables.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, load the *orderdetails* table in your data model by doing the following:

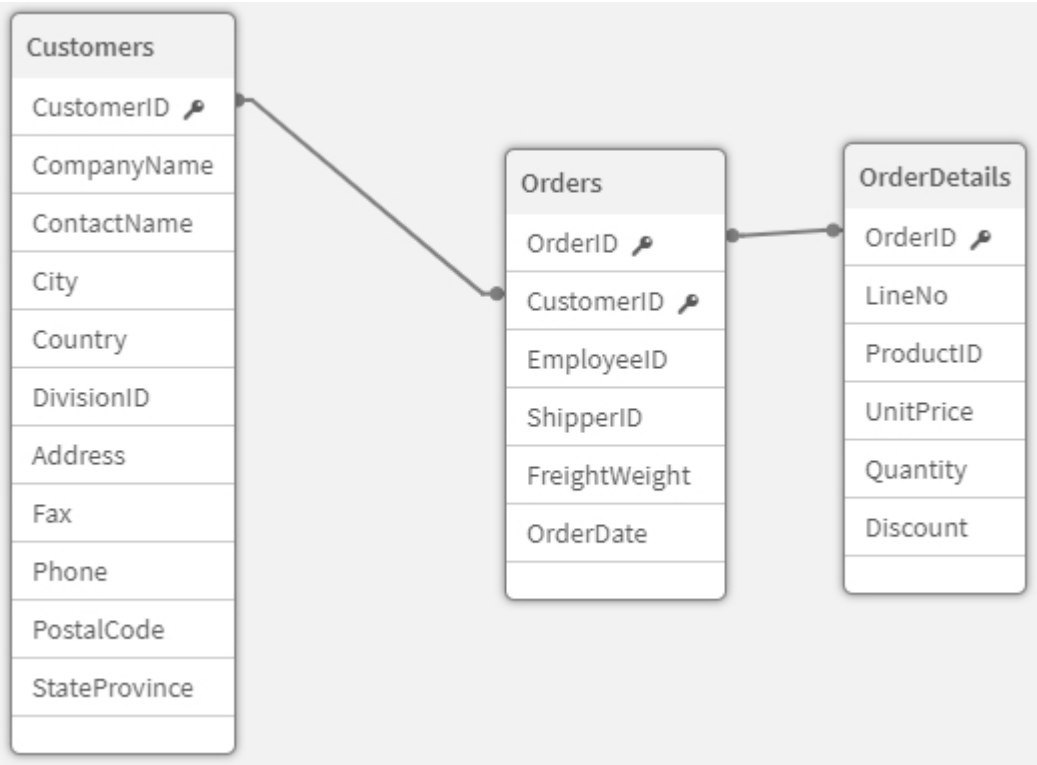
- Navigate to the end of the **Orders** script section in the **Data load editor**.
- Use the correct data connection to select data from the **ABC_2025** database.
- Load all the fields from the *orderdetails* table.
- Capitalize and move the table label *OrderDetails*: above the **LOAD** statement.
- Remove the superfluous **LIB CONNECT TO** **'ABC'**; statement, as the connection is still open from loading the *Orders* table.
- Navigate to the **Customers** script section and use the **ABC** data connection.
- Load all the fields from the *customers* table.
- Capitalize and move the table label *Customers:* above the **LOAD** statement.
- Click the **Load data** button.
- Verify that the three tables are associated, and explain why.

Note: The objective of this **challenge** is to test whether you feel comfortable with the concepts that have been discussed so far in your class. There is no **Step-by-Step** section in this activity, so you must only rely on the **Hints** below to understand the tasks you have to execute.
As you are still new to many of these concepts, please don't hesitate to reach out for more guidance (help file, instructor, Qlik Community).

Hints

- The **Select data** dialog in the data connection can be used to load the tables.
- Although both *orderdetails* and *customers* tables can be selected at the same time in the **Data load editor**, we want the two loading scripts to be placed in different sections.

Solution



Activity 4: Data model viewer

Summary:

- Use the Data model viewer and investigate the various display features.
- Analyze the Preview panel and evaluate the metrics it presents.

Qlik Sense app: ABC Sales Data

Challenge

Familiarize yourself with the components of the **Data model viewer**. Create an easier-to-read data model by exploring the following features, and be prepared to explain what they do:

- Internal/Source table views
- Collapse/Expand all options
- Preview panel
- Show linked fields

Then, answer the following questions:

1. How many **rows** exist in the *Orders* table?
2. How many **key fields** does the *Orders* table have?
3. How many **present distinct values** exist for *CustomerID* in the *Customer* table?
4. How many **present distinct values** exist for *CustomerID* in the *Orders* table? What is its **subset ratio**?
5. What is the **density** for *StateProvince* in the *Customers* table? Examine the data to explain why this is so low.

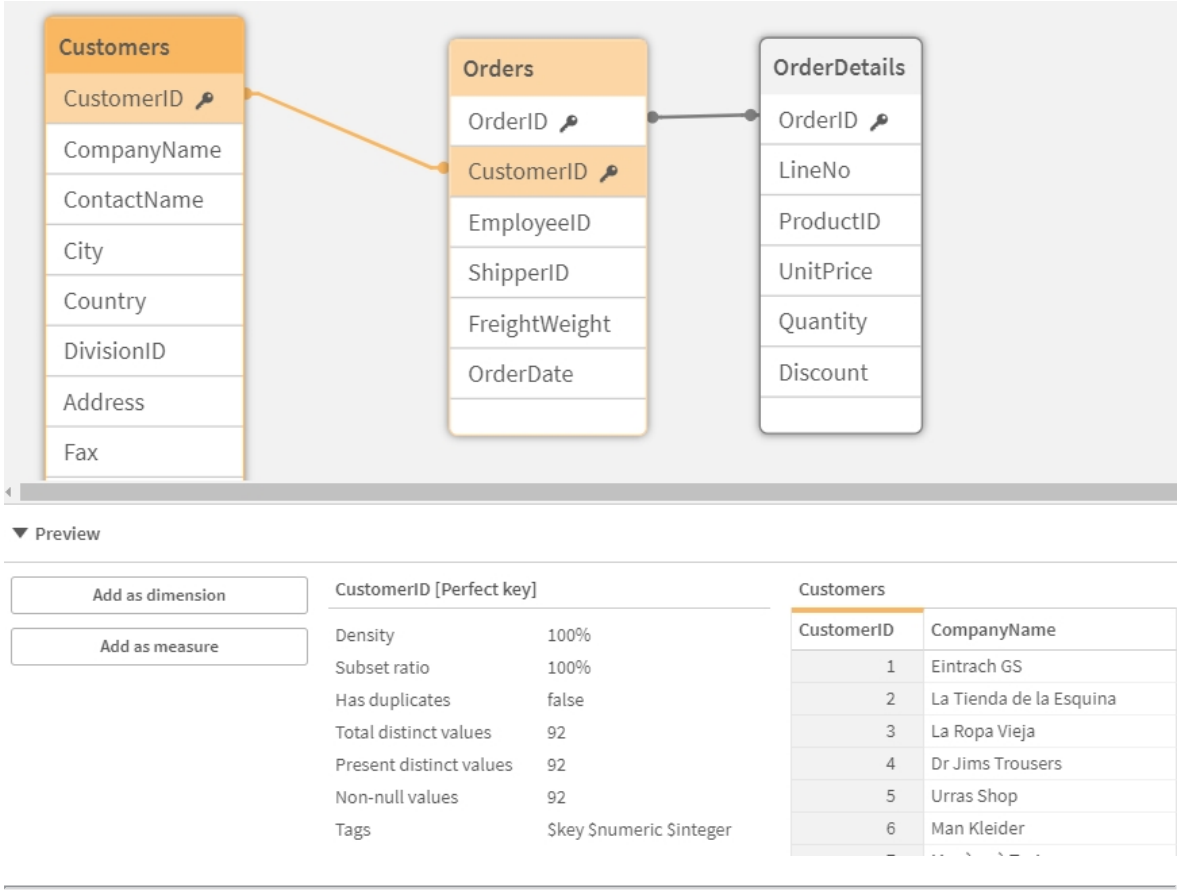
Hints

- The **Data model viewer** can be accessed from the **Prepare** drop-down menu in the top toolbar.
- The answers to the above questions are related to the **Preview** panel.

Step-by-step guide

1. In the **Data model viewer**, enable the **Internal table view**.
2. Toggle through the table views by using the **Collapse all, Show linked fields**, and **Expand all** buttons in the toolbar to see the different behavior.
3. Drag the tables to reposition them so that no tables overlap. This makes the data model easier to read.
4. Select the *Orders* table and preview its data by showing the **Preview** panel.
5. In the **Preview** panel, determine the number of **Rows** and keys.
6. In the *Customers* table, select the *CustomerID* field.
7. Use the **Preview** panel to identify how many **Present distinct** and **Total distinct values** it contains.
8. Select the *CustomerID* field in the *Orders* table to identify the number of **Present distinct** and **Total distinct values**.
9. Within the same selection, check the **Subset ratio** for the selected field.
10. Select the *StateProvince* field in the *Customers* table.
11. Check the **Density** of this field in the **Preview** panel.
12. While remaining in the **Preview** panel, examine the data using the *StateProvince* column. What do you find here and how does that explain the density?

Solution



Activity 5: Load Excel data

Summary:

- Import data from two Excel files.
- Script the loading of two tables, ensuring correct associations.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, load two new tables from two Excel files by doing the following:

- Create a new script section, *Employees*, after the **Customers** section.
- Use the **DataFiles** connection to import *employees.xlsx* and *team.xlsx* into your cloud tenant.

> **Qlik Sense on Windows:** Create a new data connection to the **ExcelFiles** folder, found within **C:\Qlik Sense Data**, and name it *ExcelFiles*.

- Load the following tables in the **Employees** section:
  - *Employees*, from the **employees** sheet in the *employees.xlsx* file. Rename the field *EmpID* to *EmployeeID* to associate the table with your data model.
  - *EmployeeTeam*, from the **employee_team** sheet in the *team.xlsx* file. Rename the field *EmpID* to *EmployeeID* for the same reason as above.
  - *Team*, from the **team** sheet in the *team.xlsx* file.
  - Add comments to all the tables.
- Load your data and verify that the tables have been added and are associated with the existing data model.

Hints

- The *EmpID* can be renamed to *EmployeeID* from either the script editor or the **Select data** dialog.
- When the **Select data** dialog appears, the following must be selected:
  - **File format**: *Excel (.XLSX)*
  - **Field names**: *Embedded field names*
  - **Header size**: *0*

Step-by-step guide

1. In the **Data load editor**, create a new script section after the **Customers** section and name it *Employees*.
2. From the **Data connections** panel, use **Select data** on the **DataFiles** connection.
3. Sourcing the data from the student file collateral, drag the *employees.xlsx* onto the **Select file** drop zone. Do the same with the *team.xlsx* file.

   Qlik Sense on Windows: Create a new data connection to the **ExcelFiles** folder, found within **C:\Qlik Sense Data**, and name it *ExcelFiles*.
   Use the **Select data** option from the new connection, to launch the **Select file** dialog.

4. With the **Select file** dialog still open, select the *employees.xlsx* file.
5. In the **Select data** dialog, check that the following are automatically selected:
   - **File format:** Excel (XLSX)
   - **Field names:** Embedded field names
   - **Header size:** 0
6. In the **Fields** area, rename the *EmpID* field to *EmployeeID* to associate the new table with the existing data model. You can now see that the *EmpID* has an alias to *EmployeeID* using the **AS** keyword.
7. Insert the generated script into the **Data load editor**.
8. Name the new table *Employees:* and, as good practice, add a suitable comment before the table label.
9. With your cursor placed at the end of the script section, add a few carriage returns for easier readability and use the same data connection again to add more tables.

10. Select the *team.xlsx* file and, when the **Select data** dialog appears, select both the *employee_team* and *team* tables. Note the numbers on the right of the table name show the number of selected fields.
11. In the *employee_team* table, replace the *EmpID* with *EmployeeID* and make sure the field is checked.
12. Insert the script and make sure that the script statement has been inserted.
13. Name the tables *EmployeeTeam* and *Team*, respectively, and add suitable comments before each table label. The code in your **Employees** section resembles the following:

```
// **** Load Employees table ****
Employees:
LOAD
    EmpID AS EmployeeID,
    "First Name",
    "Last Name",
    DateOfBirth,
    StartDate,
    EndDate,
    OfficeID,
    JobTitle
FROM [lib://DataFiles/employees.xlsx]
(ooxml, embedded labels, table is Employees);

// **** Load EmployeeTeam table ****
EmployeeTeam:
LOAD
    EmpID AS EmployeeID,
    TeamID
FROM [lib://DataFiles/team.xlsx]
(ooxml, embedded labels, table is employee_team);


// **** Load Team table ****
Team:
LOAD
    TeamID,
    Team
FROM [lib://DataFiles/team.xlsx]
(ooxml, embedded labels, table is team);
```

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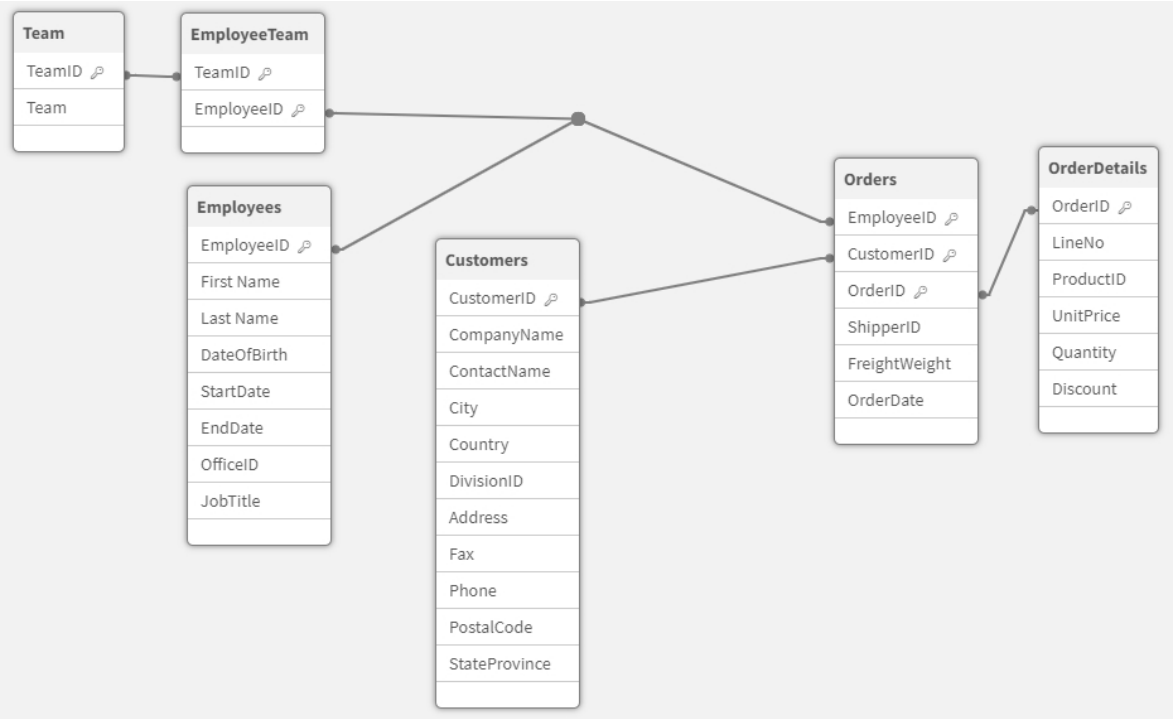
14. Click the **Load data** button and check your data model in the **Data model viewer**.

Solution



Activity 6: Load CSV data

Summary:

- Import data from a .csv file.
- Configure the import process to ensure it adheres to the file format.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, load a new table from a .csv file by doing the following:

- Navigate to the end of the **Employees** script section.
- Use the **DataFiles** connection to import *office.csv* into your cloud tenant.

  Qlik Sense on Windows: Create a new data connection to the **TextFiles** folder, found within **C:\Qlik Sense Data**, and name it *TextFiles*.

- Insert the script for the *offices* table from the *office.csv* file.
- Load your data and verify that the table has been added and is associated with the existing data model.

Hints

- When the **Select data** dialog appears, the following must be selected:
  - **File format**: *Delimited*
  - **Field names**: *Embedded field names*
  - **Header size**: *0*
  - **Delimiter**: *Comma*

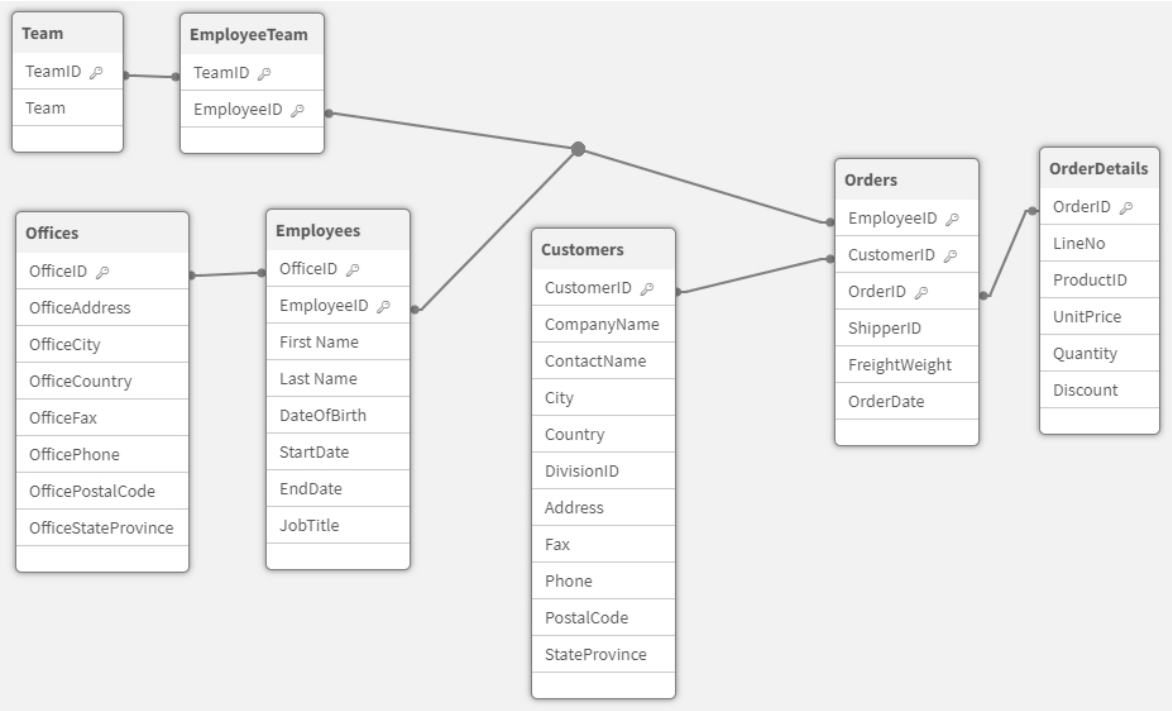
Step-by-step guide

1. Navigate to the bottom of the **Employees** script section and add a few carriage returns.
2. From the **Data connections** panel, use **Select data** on the **DataFiles** connection.
3. Sourcing the data from the student file collateral, drag the *office.csv* file onto the drop zone to import it.

   Qlik Sense on Windows: Create a new data connection to the **TextFiles** folder, found within **C:\Qlik Sense Data**, and name it *TextFiles*.
   Use the **Select data** option from the new connection, to launch the **Select file** dialog.

4. With the **Select file** dialog still open, select the *office.csv* file.
5. When the **Select data** dialog appears, notice a slightly different layout from the one for Excel files. Check that the following have automatically been selected:
   - **File format:** *Delimited*
   - **Field names:** *Embedded field names*
   - **Header size:** *0*
   - **Delimiter:** *Comma*
6. Click the **Insert script** button and, above the **LOAD** statement, name the new table *Offices*.
7. Add a suitable comment before the table label.
8. Click the **Load data** button and check your data model in the **Data model viewer**.

Solution



Activity 7 - Optional: Load fixed record data

Summary:

- Load data from a fixed record text file.
- Alias two fields and use the Data model viewer to check for correct association.

Qlik Sense app: ABC Sales Data

Challenge

In the **ABC Sales Data** app, load a new table from a fixed record .txt file by doing the following:

- Navigate to the end of the **Employees** script section.
- Use the **DataFiles** connection to import *org_structure.txt* into your cloud tenant.

  Qlik Sense on Windows: Use the **TextFiles** connection you created previously.

- Change the **File format** to *Fixed record* and use **Field breaks** to define the table columns.
- Add breaks for columns 9, 21 and 33 and preview the new table format.
- Rename *EmpID* to *EmployeeID*, *ReportsTo* to *ManagerID* and deselect the fields *StartDate* and *EndDate*.
- Name the new table *OrgStructure* and add a comment.

- Load your data and verify that the table has been added and is associated with the existing data model.

Hints

- When the **Select data** dialog appears, the following must be selected:
  - **File format**: *Fixed record*
  - **Field names**: *Embedded field names*
  - **Field break positions**: *1*
  - **Header size**: *0*
- When selecting data from a fixed record .txt file, the field breaks need to be set. They define the columns of the table.

Step-by-step guide

1. In the **Data load editor**, navigate to the end of the **Employees** script section and add a few carriage returns.
2. From the **Data connections** panel, use **Select data** on the **DataFiles** connection.
3. Sourcing the data from the student file collateral, drag the *org_structure.txt* file onto the drop zone to import it.

   Qlik Sense on Windows: Use the **TextFiles** connection you created previously.

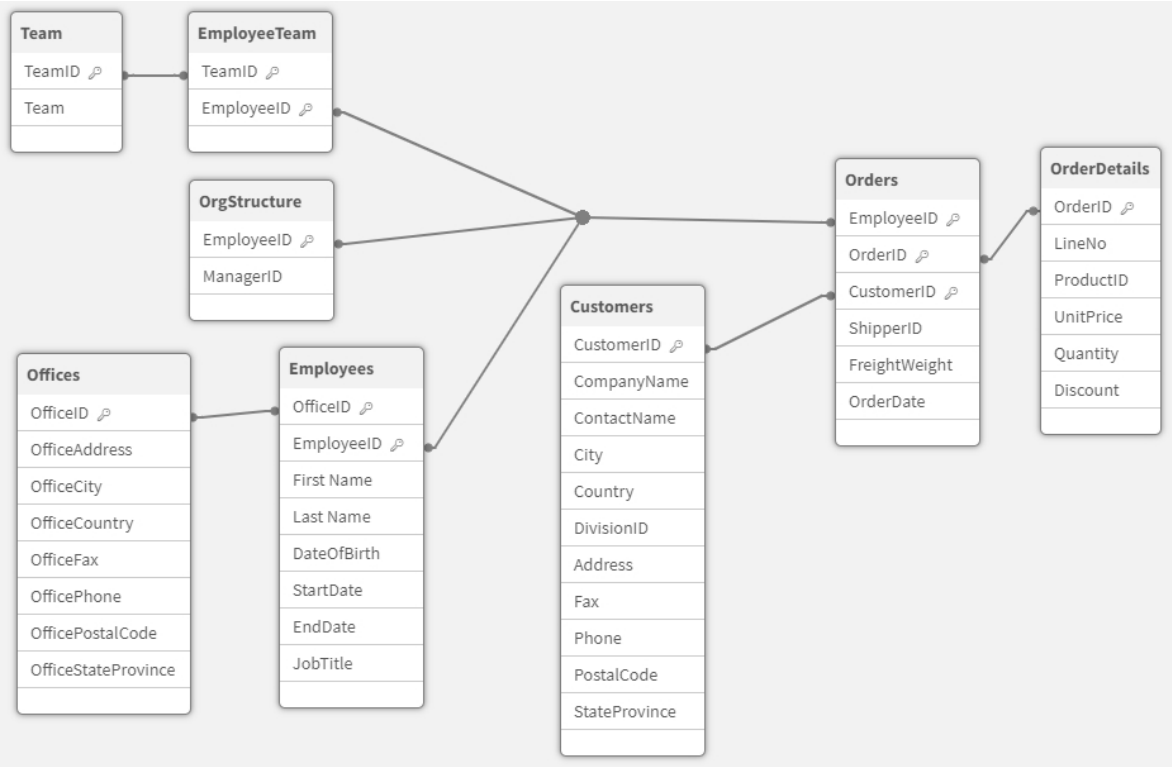
4. Select *org_structure.txt* as your data file.
5. When the **Select data** dialog appears, change the **File format** to *Fixed record*.
6. Select the **Fields breaks** view.
7. Define the fields by clicking at the start of each field column to add breaks: Click to add a delimiter at columns 9, 21, and 33. If you make a mistake, you can remove a field break by moving to the position of the field break again. When it turns red, click it again to remove it.
8. Change back to **Data preview** and verify that your table now has four columns.
9. Although you set the field breaks for *StartDate* and *EndDate* in this example, you do not want to load them. Deselect the *StartDate* and *EndDate* fields.
10. Change the following field names:
    - *EmpID* to *EmployeeID*
    - *ReportsTo* to *ManagerID*
11. Click the **Insert script** button and, above the **LOAD** statement, name the new table *OrgStructure*.
12. Add a suitable comment before the table label. Your new script should resemble the following:

```
            // **** Load OrgStructure table ****

[OrgStructure]:
LOAD
    "@1:8" AS EmployeeID,
```

```
    "@9:20" AS ManagerID
FROM [lib://DataFiles/org_structure.txt]
(fix, codepage is 28591, embedded labels);
```

Copy to clipboard

13. Click the **Load data** button and check your data model in the **Data model viewer**.

Solution